WETLAND ASSESSMENT

For the Establishment of the Nyandungu Wetlands, Ponds and its Associated Hydrological Infrastructure, Kigali, Rwanda

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DRAFT REPORT

Acronyms

CVB	Channeled Valley Bottom
EAP	Environmental Assessment Practitioner
ECO	Environmental Control Officer
EIS	Ecological Importance & Sensitivity
GIS	Geographical Information Systems
HGM	Hydro-Geomorphic
IAPs	Invasive Alien Plants
PES	Present Ecological State

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1. INTRODUCTION

1.1 Project Background and Description of the Activity

As part of a wetland rehabilitation for the construction of the Nyandungu wetland eco-park, a hydrological assessment is required. This is in part due to the presence of watercourse features and the nature of the development. A substantial area of the wetland was historically channelled and drained to allow for a greater area of arable land. Since the removal of this practice, some of the natural flora and fauna have returned. However, the hydrological regime is still highly transformed. A series of dams has been proposed in order to restore some of the functioning of this system, promote the return of biodiversity and increase the wetland area towards its original state. However, there is a concern that the dams will reduce the sustainable yield to downstream water users. Furthermore, the dams need to be interrogated to ensure that they are the optimum size to ensure that they are not overdesigned thereby wasting resources.

The primary objective of this hydrological component (which feeds into the overall rehabilitation component) is to re-instate more natural water distribution and retention patterns that will improve the overall functioning of the wetland and associated habitat.

This may involve:

- Plugging drainage channels created by historical agricultural practices to drain wetland areas for other land use purposes;
- Constructing dams in practical areas to redistribute the flow, promote the return of biodiversity and attenuate floods as the wetland would have done prior to its transformation;
- Removing invasive alien or undesirable plant species from wetlands and their immediate catchments (in conjunction with the botanical landscaping).

See the locality map in figure 1 for the proposed layout of the dams and landscaping.

Uninformed and poorly planned infrastructural developments in the vicinity of water resources, such as surface water, can rapidly degrade these resources. Thus, pre-development (or in some cases post development) assessments are required to gain an understanding of the natural environment and guide the developmental process in order that site-specific mitigation measures can be put in place.

Figure 1 Typical rehabilitation intervention measures on site

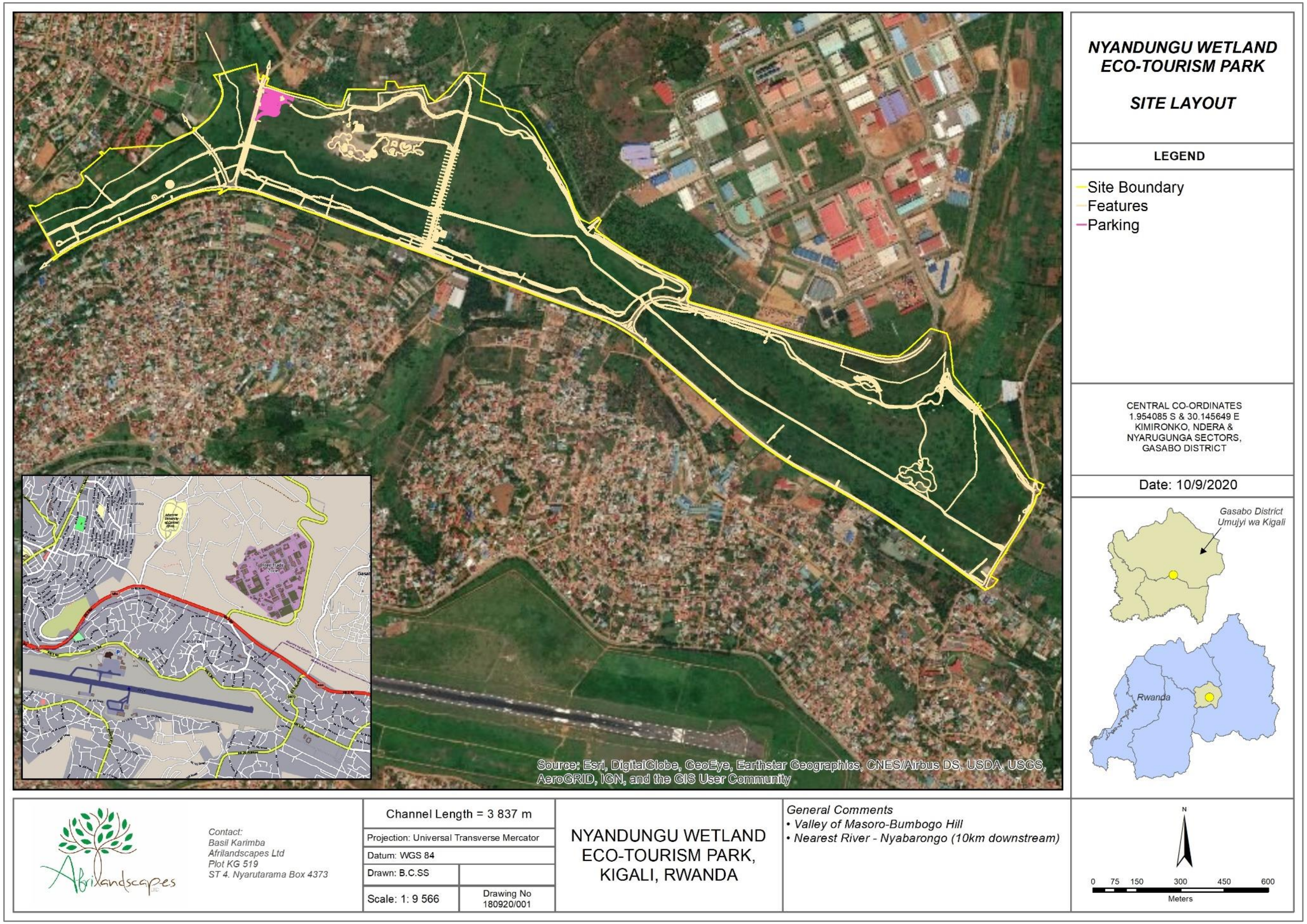

Figure 2 Locality map of Nyandungu wetland park

1.2 Terms of reference

<u>Hydrological Assessment</u>

To characterise the baseline surface water regime at the site and the catchments. The study will identify risks posed to surface water and provide inputs to management plans and designs. The scope of work includes:

i. **General Hydrological:**

- Site and catchment assessment (surface water details) for mapping;
- Assessment of rainfall data;
- Assessment of interventions – gabions, dams, channel rehabilitation, landscaping and civil works; and
- Assessment of topographical report and designs.

ii. **Surface Water & Flood Hydrology:**

- Determine stream flows, flood magnitudes and floodlines with the interventions of watercourses on-site by:
  - Hydrological assessment, undertaken by the:
    - Analysis of rainfall data available;
    - Determination of the catchment characteristics;
    - Determination of the Manning's n-values; and
    - Analysis of the river reach network
    - Estimation of the design flood;
  - Hydraulic analysis, illustrated by the:
    - Compilation of the river reach model and flood line using HEC-RAS and HEC-geoRAS;
    - Determination of the flood risk and flood hazard throughout the study site; and
    - Recommendation of mitigation options associated with the hydraulic analysis.
  - Consolidate results in a report with:
    - Flood line maps; and
    - A final flood line report.

- Recommendations for future safety measures.

iii. **Wetland Health/Condition Assessment**

The condition/Present Ecological State (PES) of the delineated riverine and wetland areas present within all sectors of the site; as well as the functional importance of any wetlands present within and near the development footprint would be assessed. This will involve:

a. an assessment of the delineated riverine areas by:
   i. determining the condition/PES of the riverine system using the rapid/qualitative Index of Habitat Integrity (IHI) tool (Kleynhans, 1996) for rivers (in-stream and riparian habitats assessed separately); and
   ii. determining the health/ecological importance & sensitivity (EIS) using the DWAF riverine EIS tool (Kleynhans, 1999).
b. an assessment of the delineated wetland areas by:
   i. determining the condition/ PES of the delineated wetlands using the Level 1 WET-Health tool (Macfarlane et al, 2009); and
   ii. determining the ecological importance & sensitivity (EIS) of the delineated wetlands using the Department of Water Affairs and Forestry (DWAF) wetland EIS tool (Duthie, 1999).
c. an impact assessment to investigate, evaluate and assess the impacts of the abovementioned activities on the environment.

iv. Recommendations:

- Provide an overall report on the state of the project (including all interventions/developments);
- Provide design extent of flood areas;
- Provide an assessment of the state of the wetlands in contrast to the pre-development state;
- Provide a surface water monitoring plan.

1.3 Classification System for Wetlands and Other Aquatic Systems

Differences in terminology can lead to confusion in the scientific and consulting fields. As such, terminology used in the context of this report needs to be defined. A watercourse, wetland and riparian habitat as follows:

- A **watercourse** means - (a) a river or spring; (b) a natural channel in which water flows regularly or intermittently; (c) a wetland, lake or dam into which, or from which, water flows;
- A **wetland** means land which is transitional between terrestrial and aquatic systems where the water table is usually at or near the surface, or the land is periodically covered with shallow water, and which land in normal circumstances supports or would support vegetation typically adapted to life in saturated soil.
- A **riparian habitat** includes the physical structure and associated vegetation of the areas associated with a watercourse which are commonly characterised by alluvial soils, and which are inundated or flooded to an extent and with a frequency sufficient to support vegetation of species with a composition and physical structure distinct from those of adjacent land areas.

Any features meeting these criteria within the development site were delineated and classified Inland wetland systems (non-coastal) are ecosystems that have no existing connection to the ocean which are inundated or saturated with water, either permanently or periodically (Ollis *et. al*., 2013). The level 3 classification has been divided into four landscape units. These are:

a) **Slope** – located on the side of a mountain, hill or valley that is steeper than lowland or upland floodplain zones.
b) **Valley Floor** – gently sloping lowest surface of a valley, excluding mountain headwater zones.
c) **Plain** – extensive area of low relief. Different from valley floors in that they do not lie between two side slopes, characteristic of lowland or upland floodplains**.**
d) **Bench** (hilltop/saddle/shelf) - an area of mostly level or nearly level high ground, including hilltops/crests, saddles and shelves/terraces/ledges.

Level 4 HGM types (which is commonly used to describe a specific wetland type) have been divided into 8 units. These are described as follows:

- **Channel** (river, including the banks) - an open conduit with clearly defined margins that (i) continuously or periodically contains flowing water. Dominant water sources include concentrated surface flow from upstream channels and tributaries, diffuse surface flow or interflow, and/or groundwater flow.
- **Channelled valley-bottom wetland** - a mostly flat valley-bottom wetland dissected by and typically elevated above a channel (see channel). Dominant water inputs to these areas are typically from the channel, either as surface flow resulting from overtopping of the channel bank/s or as interflow, or from adjacent valley-side slopes (as overland flow or interflow).
- **Un-channelled valley-bottom wetland** - a mostly flat valley-bottom wetland area without a major channel running through it, characterised by an absence of distinct channel banks and the prevalence of diffuse flows, even during and after high rainfall events.
- **Floodplain wetland** - the mostly flat or gently sloping wetland area adjacent to and formed by a Lowland or Upland Floodplain river, and subject to periodic inundation by overtopping of the channel bank.
- **Depression** - a landform with closed elevation contours that increases in depth from the perimeter to a central area of greatest depth, and within which water typically accumulates. Dominant water sources are precipitation, ground water discharge, interflow and (diffuse or concentrated) overland flow.
- **Flat** - a near-level wetland area (i.e. with little or no relief) with little or no gradient, situated on a plain or a bench in terms of landscape setting. The primary source of water is precipitation.
- **Hillslope seep** - a wetland area located on (gentle to steep) sloping land, which is dominated by the colluvial (i.e. gravity-driven), unidirectional movement of material down-slope.

- **Valley head seep** - a gently-sloping, typically concave wetland area located on a valley floor at the head of a drainage line, with water inputs mainly from subsurface flow.

2. STUDY SITE

2.1 General Description

The Nyandungu Wetland is located in the valley of Masoro-Bumbogo hill in Ndera, Gasabo District and Kanombe-Nyarugunga hill in Kicukiro District (Gasabo 3D & Astrik Int, 2017). The wetland park is divided into five Sectors, starting below La Palisse Hotel and ending at K9 Avenue. The site runs parallel to Kigali International Airport (cf. Figure 1).

A significant amount of work has been undertaken on site. This includes:

- Rehabilitation of the incised channel through gabion structures, channel shaping and re-vegetation;
- Construction of numerous dams/ponds that encourage bird life and attenuate peak flood events;
- The upgrade of various road culverts throughout the site;
- The construction of stormwater drains surrounding the site;
- The construction of various tourism related structures; and
- The complete revegetation of the site.

The area is dominated by grassland and wetland sedge species. The landscape is that of a typical valley bottom but is interrupted by numerous road crossings as well as being encroached by residential, commercial and industrial areas. Rainfall occurs throughout the year, with mean annual precipitation being approximately 1158 mm. Much precipitation comes in the form of thunderstorms and mist is uncommon. Summers and winters are warm to hot.

Table 1 Mean monthly rainfall and temperature observed at Kimironko (derived from historical data)

	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	Ann
Mean Rainfall (mm)	99.9	99.9	121.8	197.0	107.0	17.7	7.5	63.1	89.5	104.7	143.9	106.9	1158
Mean Temperature (°C)	20.6	20.9	19.8	18.4	18.3	19.0	20.0	21.1	21.5	20.4	19.3	19.7	19.9

Figure 3 The typical features throughout the wetland park

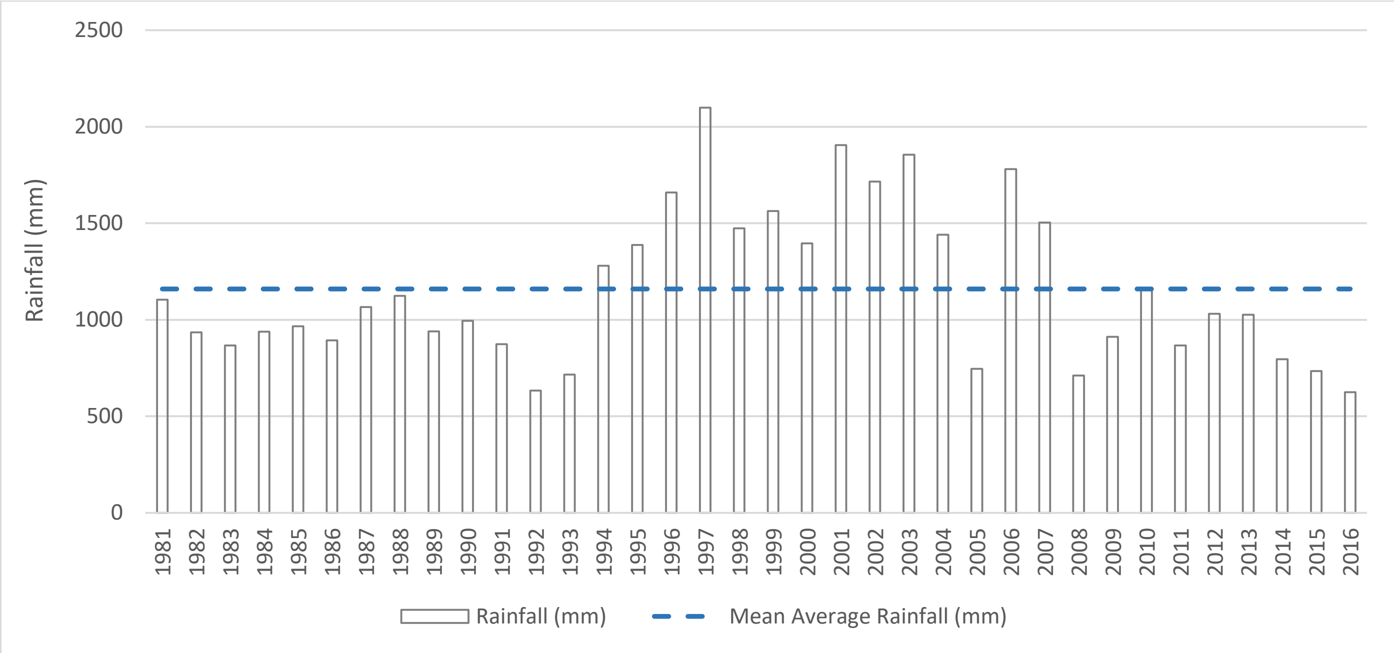

Figure 4 Long-term rainfall near the wetland park

3. METHODOLOGY

A detailed description of the methods has been provided. The regional context and desktop analysis were used as the point of departure. Subsequently, numerous site visits were undertaken to delineate any wetlands and riparian areas and their change through the project phases. These systems were then assessed to determine the potential impacts that have been caused. The assessment of these systems considered the following tools where relevant:

Table 2 Assessment approach and the recommended tools for rivers and wetlands

Aquatic Component	Method/Technique	Tool Utilized
Rivers	Delineation	*A Practical Field Procedure for Identification and Delineation of Wetland and Riparian Areas' (DWAF, 2005).*
	Classification	*Wetland Classification System for Wetlands and other Aquatic Ecosystems (Ollis et al, 2014).*
	River condition/Present Ecological State (PES)	DWAF IHI (Index of Habitat Integrity) tool (Kleynhans, 1996) for rivers (riparian habitat only)
Wetlands	Delineation	*A Practical Field Procedure for Identification and Delineation of Wetland and Riparian Areas' (DWAF, 2005).*
	Classification	*Wetland Classification System for Wetlands and other Aquatic Ecosystems (Ollis et al, 2014).*
	Wetland condition/Present Ecological State (PES)	Level 1 WET-Health tool (Macfarlane *et al*., 2009)
	Wetland Functional/Ecosystem Services Assessment	Level 2 WET-EcoServices assessment tool (Kotze *et al*., 2009)
	Wetland Ecological Importance & Sensitivity (EIS)	DWAF wetland EIS tool (Duthie, 1999)

Table 3 Data type and source for the Bergview assessment

Data Type	Year	Source/Reference
Aerial/Satellite Imagery	2016-2020	Surveyor General, LandSat
Topographical	2019	AfriLandscapes
River Shapefile	2019	User Defined
Land Cover	2019	User Defined
CAD Drawings	2019/2020	AfriLandscapes

*Data will be provided on request

3.1 Regional Context

3.1.1 Terrain

Terrain data was provided by the client and AfriLandscapes.

3.1.2 Soils

On site soil surveys were undertaken.

3.1.3 Vegetation

The vegetation component has been reported on separately (Starke, ongoing).

3.2 Extent, Classification and Habitat Characteristics

The boundary of wetlands and riparian areas occurring on the site was identified and delineated according to the globally recognised technique documented by the Department of Water Affairs wetland delineation manual '*A Practical Field Procedure for Identification and Delineation of Wetland and Riparian Areas*' (Department of Water Affairs, 2005). Land cover data, contour data and the latest aerial imagery were examined in a thorough desktop analysis of the site. This provided important background information to the specialists' understanding of the broader context of the landscape (e.g. baseline vegetation, geology and climate). An on-site delineation was undertaken as described below.

3.2.1 Wetland Delineation

The following indicators stipulated in the national delineation guidelines were considered in the field. Not necessarily all of these indicators were used at each site. Mention was made in the results which of these indicators were used:

- **Terrain Unit Indicator** – this relates to the position within the landscape where a wetland may occur. A typical landscape can be divided into five main terrain units, namely the crest (hilltop), scarp (cliff), midslope (often a convex slope), footslope (often a concave slope), and valley bottom. As wetlands occur where there is a prolonged presence of water, the most common place one would expect to find wetlands is on the valley bottom (Rountree *et al,* 2008*)*.

- **Soil Form Indicator** – this identifies the soil forms, which are associated with prolonged and frequent saturation.

- **Soil Wetness Indicator** - Prolonged saturation of soil results in the development of anaerobic conditions, which has a characteristic effect on soil morphology, causing two important redoximorphic features: mottling and gleying. The hue, value and chroma of soil samples obtained at varying depths can be visually interpreted with the aid of the Munsell Colour Chart and the interface between wetland and non-wetland zones determined.

- **Vegetation Indicator** – Plant species have varying tolerances to different moisture regimes. The presence, composition and distribution of specific hydrophytic plants within a system can be used as an indication of wetness and allow for inference of wetland characteristics.

The area was extensively traversed, auger sample points were taken as required and the exact location of sample points logged using a Garmin GPSMAP 64. At each sampling point the soils were sampled at depths of 0-10 cm and 40-50 cm below surface. The soil value, hue and matrix chroma were recorded for each sample according to the Munsell Soil Colour Chart, and the degree of mottling and/or presence of concretions were recorded. Although the site was severely transformed, any vegetation of interest was noted for the assessments. If the author was not able to identify any potentially important species, a leaf and bark sample was taken for analysis using a key guide. A separate pedological survey was also undertaken by Gasabo 3D in 2017.

3.2.2 Riparian Delineation

Riparian area/zone delineation is similar to wetland delineation in that indicators are used to define the edge of the system. It considers indicators such as topography, vegetation, alluvial soils, and deposition of material to mark the outer edge of the macro-channel and its associated vegetation. The Figure 5 shows the typical morphology of a river channel.

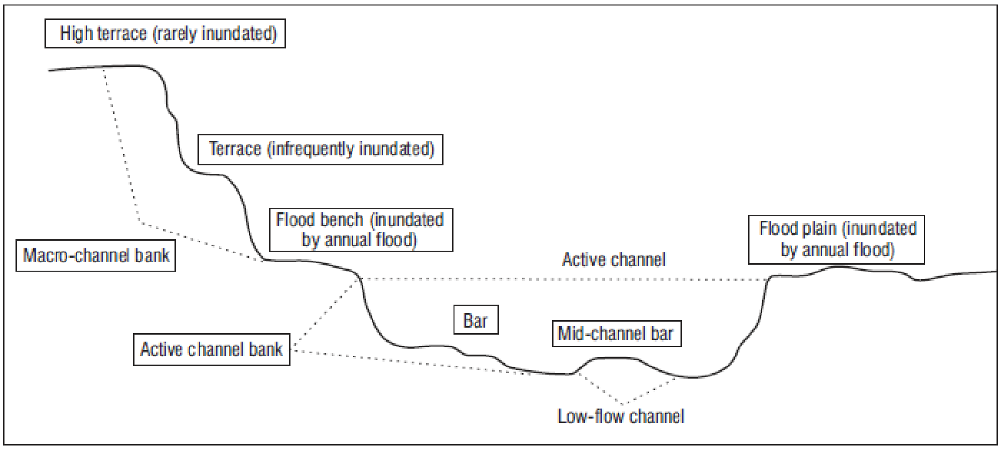

Figure 5 **Typical cross-section of a river showing channel morphology 'A Practical Field Procedure for Identification and Delineation of Wetland and Riparian Areas – Edition 1' (Department of Water Affairs, 2005)**

A Practical Field Procedure for Identification and Delineation of Wetland and Riparian Areas (DWAF, 2005) was used in the delineation of the riparian zone boundary. Delineated riparian zones were then classified using an HGM classification system based on the system proposed by Ollis *(*2013*)*. According to Cowan *et al*. (2005), riparian ecosystems are separated from other wetland ecosystems on the following three major features:

1. They have linear form as a consequence of their proximity to rivers and form a boundary between the terrestrial and aquatic ecosystems.
2. Energy and materials from the surrounding landscape converge and pass through riparian ecosystems. This amount is greater in terms of unit area than with any other system.
3. Riparian ecosystems are connected hydrologically to both upstream and downstream ecosystems (intermittently).

An example of the soil sampling approach is provided in Figure 6.

Figure 6 **Soil sampling undertaken at the site**

3.3 Present Ecological State (PES) Assessment for Riparian Areas

3.3.1 Present Ecological State (adapted from WET-Health, Macfarlane *et al*., 2008)

A WET-Health (Macfarlane *et al*., 2009) Level 1 Rapid Appraisal was used to assess the eco-physical health of any wetlands in the study area. Focusing on geomorphology, hydrology and vegetation, the tool examines the impacts and indicators of change within the system and its catchment by determining the deviation (in terms of structure and function) from the natural reference condition. The outcomes of the appraisal place importance on issues that should be addressed through rehabilitation, mitigation and/or prevention measures. A standardized scoring system allows for consistencies between different systems and reduces user subjectivity.

Scores are allocated according to the magnitude and extent of impact. These scores are integrated to produce an overall score for Present Ecological State (PES) of the system – *namely, natural, largely natural, moderately modified, largely modified, extensively modified,* and *critically modified.*

3.3.2 Index of Habitat Integrity (IHI)

The ecological integrity of a river is defined as its ability to support and maintain a balanced, integrated composition of physico-chemical and habitat characteristics, as well as biotic components on a temporal and spatial scale that are comparable to the natural characteristics of ecosystems of the region (Kemper, 1999). The observed or deduced condition of these criteria as compared to what it could have been under unperturbed conditions is surmised to indicate a change in the habitat integrity. The methodology is based on the qualitative assessment of a number of pre-weighted criteria which indicate the integrity of the in-stream and riparian habitats available for use by riverine biota. Tables 6, 7 & 8 provide the list of criteria and their scores, the impact category and the final scores for the IHI assessment that were used in the calculations.

Table 4 Criteria used in the assessment of the habitat integrity

Criterion	Relevance
Water abstraction	Direct impact on habitat type, abundance and size. Also implicated in flow, bed, channel and water quality characteristics. Riparian vegetation may be influenced by a decrease in the supply of water.
Flow modification	Consequence of abstraction or regulation by impoundments. Changes in temporal and spatial characteristics of flow can have an impact on habitat attributes such as an increase in duration of low flow season, resulting in low availability of certain habitat types or water at the start of the breeding, flowering or growing season.
Bed modification	Regarded as the result of increased input of sediment from the catchment or a decrease in the ability of the river to transport sediment (Gordon *et al*., 1993). Indirect indications of sedimentation are stream bank and catchment erosion. Purposeful alteration of the stream bed, e.g. the removal of rapids for navigation (Hilden & Rapport, 1993) is also included.
Channel modification	May be the result of a change in flow, which may alter channel characteristics causing a change in marginal instream and riparian habitat. Purposeful channel modification to improve drainage is also included.
Water quality modification	Originates from point and diffuse point sources. Measured directly or agricultural activities, human settlements and industrial activities may indicate the likelihood of modification. Aggravated by a decrease in the volume of water during low or no flow conditions.
Inundation	Destruction of riffle, rapid and riparian zone habitat. Obstruction to the movement of aquatic fauna and influences water quality and the movement of sediments (Gordon *et al*., 1992).
Exotic macrophytes	Alteration of habitat by obstruction of flow and may influence water quality. Dependent upon the species involved and scale of infestation.
Exotic aquatic fauna	The disturbance of the stream bottom during feeding may influence the water quality and increase turbidity. Dependent upon the species involved and their abundance.
Solid waste disposal	A direct anthropogenic impact which may alter habitat structurally. Also a general indication of the misuse and mismanagement of the river.
Indigenous vegetation removal	Impairment of the buffer the vegetation forms to the movement of sediment and other catchment runoff products into the river (Gordon *et al*., 1992). Refers to physical removal for farming, firewood and overgrazing.
Exotic vegetation encroachment	Excludes natural vegetation due to vigorous growth, causing bank instability and decreasing the buffering function of the riparian zone. Allochtonous organic matter input will also be changed. Riparian zone habitat diversity is also reduced.
Bank erosion	Decrease in bank stability will cause sedimentation and possible collapse of the river bank resulting in a loss or modification of both instream and riparian habitats. Increased erosion can be the result of natural vegetation removal, overgrazing or exotic vegetation encroachment.

Table 5 Impact classes and their associated scores

Impact category	Description	Score
None	No discernible impact, or the modification is located in such a way that it has no impact on habitat quality, diversity, size and variability.	0
Small	The modification is limited to very few localities and the impact on habitat quality, diversity, size and variability is also very small.	1-5
Moderate	The modifications are present at a small number of localities and the impact on habitat quality, diversity, size and variability is also limited.	6-10
Large	The modification is generally present with a clearly detrimental impact on habitat quality, diversity, size and variability. Large areas are, however, not influenced.	11-15
Serious	The modification is frequently present and the habitat quality, diversity, size and variability in almost the whole of the defined area is affected. Only small areas are not influenced.	16-20
Critical	The modification is present overall with a high intensity. The habitat quality, diversity, size and variability in almost the whole of the defined section are influenced detrimentally.	21-25

Table 6 Description of the IHI categories.

Category	Description	Score (% of total)
A	Unmodified, natural.	100
B	Largely natural with few modifications. A small change in natural habitats and biota may have taken place but the ecosystem functions are essentially unchanged.	80-99
C	Moderately modified. A loss and change of natural habitat and biota have occurred but the basic ecosystem functions are still predominantly unchanged.	60-79
D	Largely modified. A large loss of natural habitat, biota and basic ecosystem functions have occurred.	40-59
E	The loss of natural habitat, biota and basic ecosystem functions are extensive.	20-39
F	Modifications have reached a critical level and the lotic system has been modified completely with an almost complete loss of natural habitat and biota. In the worst instances the basic ecosystem functions have been destroyed and the changes are irreversible.	0-19

3.4 Functional Assessment of Wetlands

3.4.1 Ecosystem Goods and Services (WET-EcoServices, Kotze et al, 2008)

The WET-EcoServices tool (Kotze *et al.,* 2005) allows measurement of ecosystem goods and services (eco-services) provided by a wetland system. Eco-services refer to the benefits obtained from ecosystems. These benefits may be derived from outputs that can be consumed directly; indirectly (which arise from functions or attributes occurring within the ecosystem), or possible future direct or indirect uses (Howe *et al.,* 1991).

WET-EcoServices provides structured guidelines that allow the importance of the wetland to be scored according to its ability to deliver fifteen different ecosystem services, shown below –

Table 7 Ecosystem services considered (WET-EcoServices, Kotze *et al.*, 2005)

Direct benefits	Indirect benefits
• **Cultural benefits** Cultural heritage Tourism and recreation Education and research • **Provisioning benefits** Provision of cultivated foods Provision of harvestable resources Provision of water for human use Biodiversity maintenance	• **Regulating and supporting benefits** Flood attenuation Streamflow regulation Carbon storage • **Water quality enhancement benefits** Sediment trapping Phosphate assimilation Nitrate assimilation Toxicant assimilation Erosion control

3.5 Impact Assessment

The aim of the impact assessment is to identify the impacts that the current activity, as well as the remaining construction and operational phase of the development will have on the receiving environment. If avoidance is not possible, mitigation is required in the form of practical actions (Ramsar Convention, 2008). Mitigation actions can be grouped into the following:

i. **Pre-construction:** This may take the form of changes in the scale of the development (e.g. reduce the size of the development), location of development (e.g. find an alternative area with less impact), and design (e.g. change the structural design to accommodate flows and continuity).
ii. **Construction:** This may take the form of a process change (e.g. changes in construction methods), siting (e.g. locality to sensitive areas), sequencing and phasing (e.g. construction during seasonal periods).
iii. **Operational:** This may take the form of changes in post management (e.g. change management to match unpredicted impacts), monitoring (e.g. frequent checks by an ECO), rehabilitation (e.g. if mitigation actions are not effective).

An assessment of the potential impacts of the Bergview dam activities was guided by the EKZNW handbook for biodiversity impact assessments (2011). As it is an existing impact, a pre- and post-rehabilitation assessment was undertaken.

It must be noted that am **impact assessment was undertaken** to identify pre-rehabilitation and post-rehabilitation impacts.

4. LIMITATIONS AND ASSUMPTIONS

In order to apply generalized and often rigid scientific methods or techniques to natural, dynamic environments, a number of assumptions are made. Furthermore, a number of limitations exist when assessing such complex ecological systems. The following constraints may have affected this assessment –

- A Garmin GPSMAP 64 was used in the mapping of waypoints on-site. The accuracy of the GPS is affected by the availability of corresponding satellites and accuracy ranges from 1 to 3 m after post-processing corrections have been applied.

- A Munsell Soil Colour Chart was used to assess soil morphology. This tool requires that a dry sample of soil be assessed. However, due to in-field time constraints, slightly wet soil samples were assessed. Wet samples would have consistently lower values than dry soils; and this is taken into consideration.

- Although the vegetation was taken into account, protected and threatened species, such as bulbs that have not emerged, may not have been identified.

- The sampling was undertaken over various climatic seasons. Given these circumstances, extra caution was taken to ensure that watercourse features were not overlooked.

- The irrigation of existing croplands has impacted on the hydrological properties (natural and irrigation induced waterlogging) in the landscape surrounding the site.

- There was little to no data on flows out of the system. The catchment is small and the watercourse associated with the site has been transformed due to urban expansion and farming.

5. RESULTS AND DISCUSSION

5.1 Regional Context

5.1.1 Terrain

Various topographical surveys have been undertaken for the site (Figure 7 & 8).

Figure 7 Contour lines for Sector 3

5.1.2 Soils and Geology

The Nyandungu wetland park is primarily dominated by quartzite, schist, granite, alluvia and colluvia (Rema, 2012). Nyandungu wetland is situated in a lowland area which receives water from the surrounding hills and upper stream of Mwanana stream flowing from Kimironko which also discharges sewage water from Kimironko Prison and SEFZ, Kabagenda stream flowing from Ndera area, Remera stream flowing from La Pallisse area and Nyarugunga River from Kanombe areas resulting in the development of a sediment of fossil fluvial deposits on the lowland (Rema, 2012). The dynamic of soil formation has been studied (Rugege D, 2004) and indicated that there are two main soil series developed in the Nyandungu wetland, namely; Nyamatebe and Rwotsa soils. These soils are developed from alluvial materials, their texture is heavy clay, and they are grey and present a Cambic development. They are poorly drained and much deeper which give proper function of filtration of the wetland.

The pedological survey showed that the Nyandungu wetland is characterized by clay soil at the center and silt-sand at the ridge of the wetland. The sandy soils are predominant in the granitic area on the North side of the wetland (Gasabo 3D & Astrik Int, 2017).

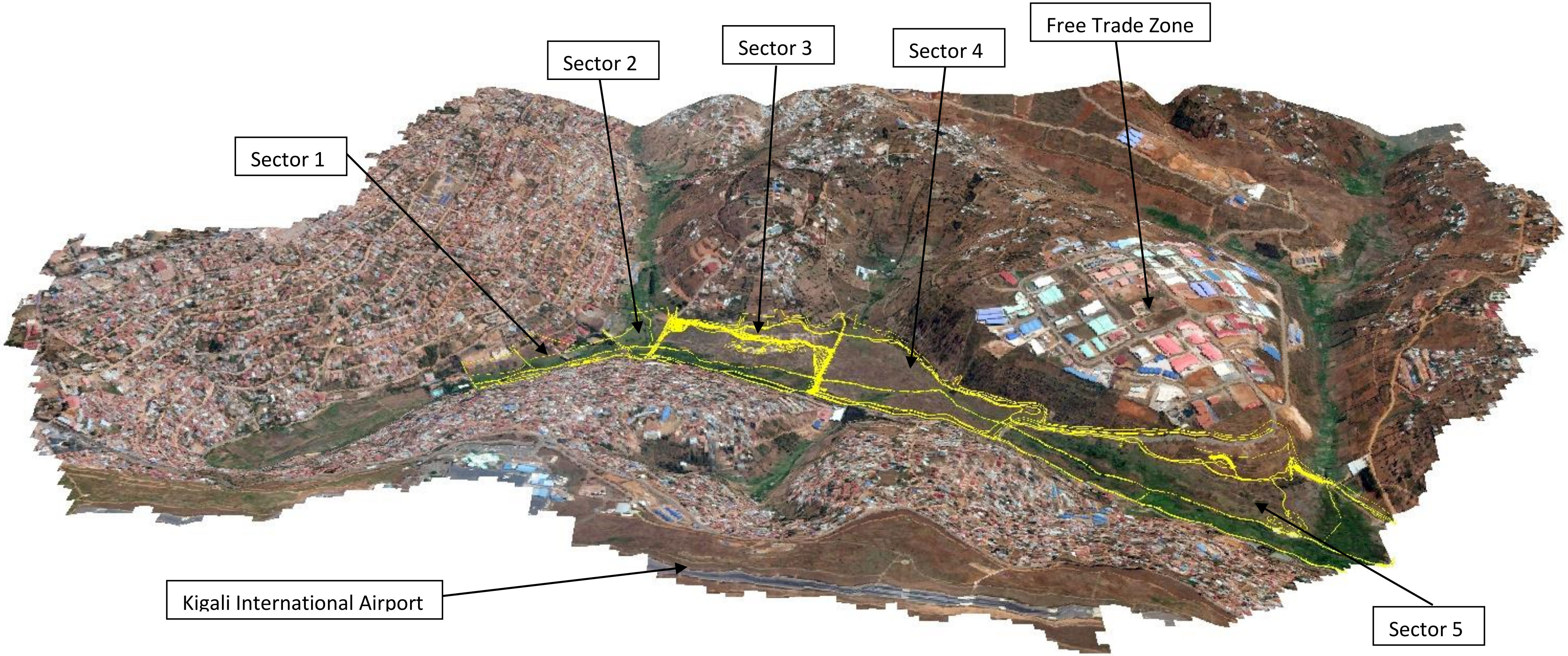

Figure 8 **Terrain model of Nyandungu wetland eco-tourism park**

5.2 Extent, Classification and Habitat Characteristics

The current land cover was obtained from the site visit. The site is surrounded by urban, commercial and industrial infrastructure. The wetland site has been historically transformed through the draining of the wetland to create more arable land. This has led to severe damage to the natural state of this site. In the last decade, sand mining has also led to the removal topsoil and clearing of natural vegetation.

The dominant species in the riparian areas were mostly indigenous sedge and grassland Fortunately, some of these areas are still intact. A subsequent revegetation programme has led to the transformation of the site to near completely indigenous.

The site consists of some areas of hydrological interest and these areas have been tabulated (Table 8) and described in detail. The HGM units are further illustrated in Figure 11. Any wetlands within Sector 1 to 5 were assessed for wetland health or functionality. The wetlands have been delineated to show no go areas and to monitor their change through the on-going rehabilitation.

Figure 9 **Typical vegetation around the site**

The following wetland systems were identified

- HGM 1: Channeled valley bottom wetland (originally un-channeled).
- HGM 2: Various artificial drainage lines.

The majority of the soils identified adjacent to the watercourse were sandy clay soils (Alluvium - pale-brown sand underlain by clay). No hydric soil characteristics were found outside of the wetland. The hydric soils, identified by gleyed or mottled characteristics, were found at a depth of 0-30 cm along the edges of the visibly clear wetlands.

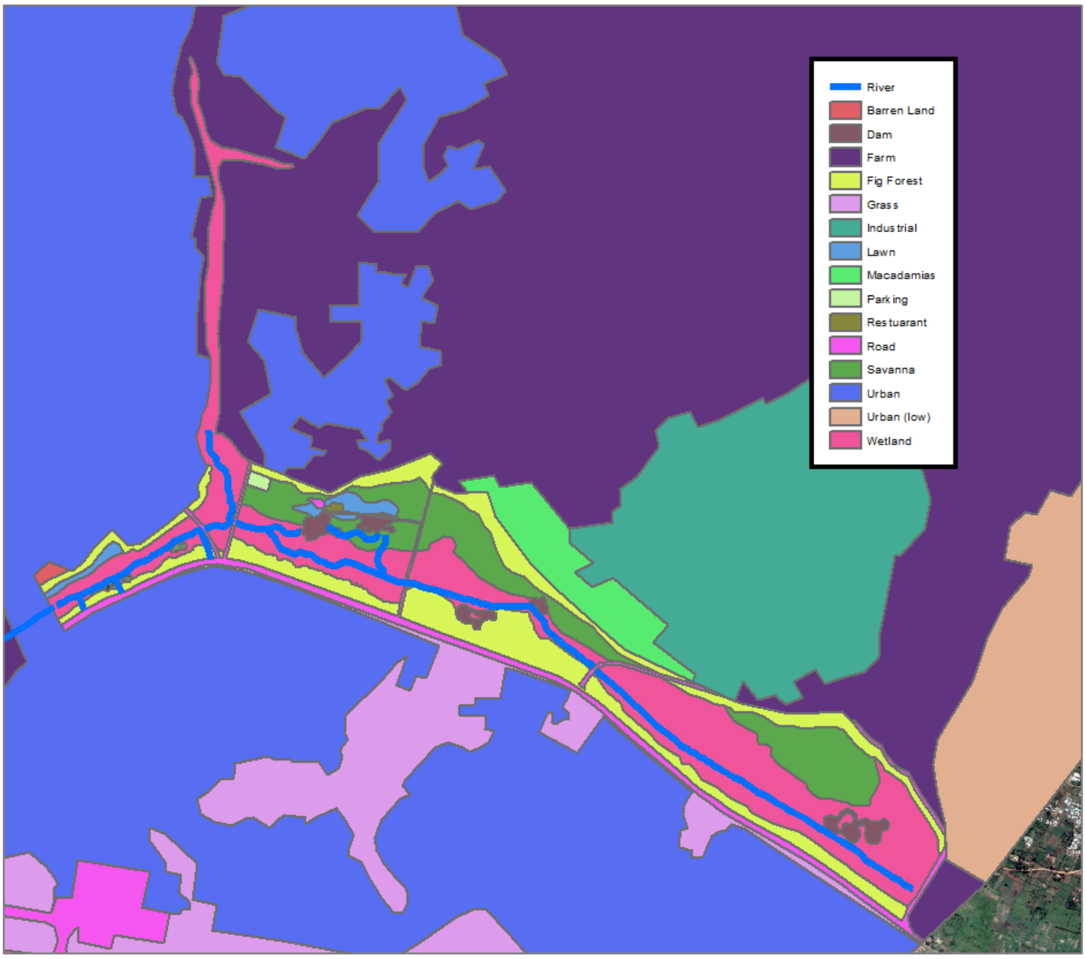

Figure 10 **Current land use around the Nyandungu wetland eco-tourism park**

Table 8 Description of HGM units

Feature	Wetland/Riparian/Artificial	Description & Vegetation	Soil Characteristics	On-site images
Channeled Valley Bottom Wetland (HGM 1)	Wetland –All Sectors connected	Valley bottom wetland system with a defined channel that merges with a larger system across the road. Colours indicate the oxidation of dissolved iron particles with some colloidal iron. Mottled clay still present. Obligate species are present.	Mottle % - 2-5% Hue – 10YR Value – 4 Chroma – 3 (Brown) Depth sampled: 0-0.5m Low Organic matter content in the upper layer	
Drainage Lines	Wetland	Various grey water drainage lines for stormwater and grey water from the extensive infrastructure surrounding the site.	N/A	

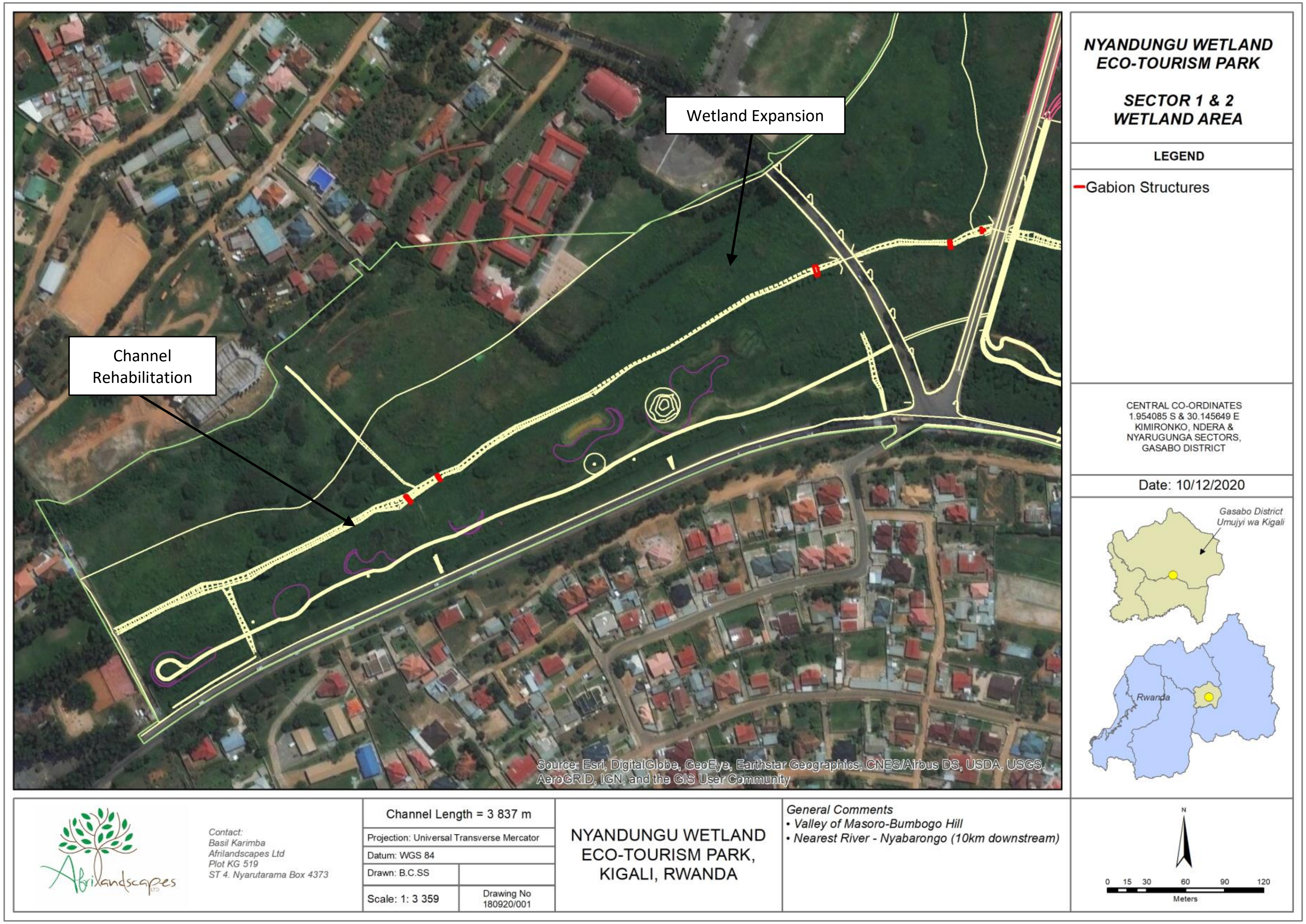

Figure 11 Wetland area of Sector 1 & 2 of Nyandungu wetland eco-tourism park

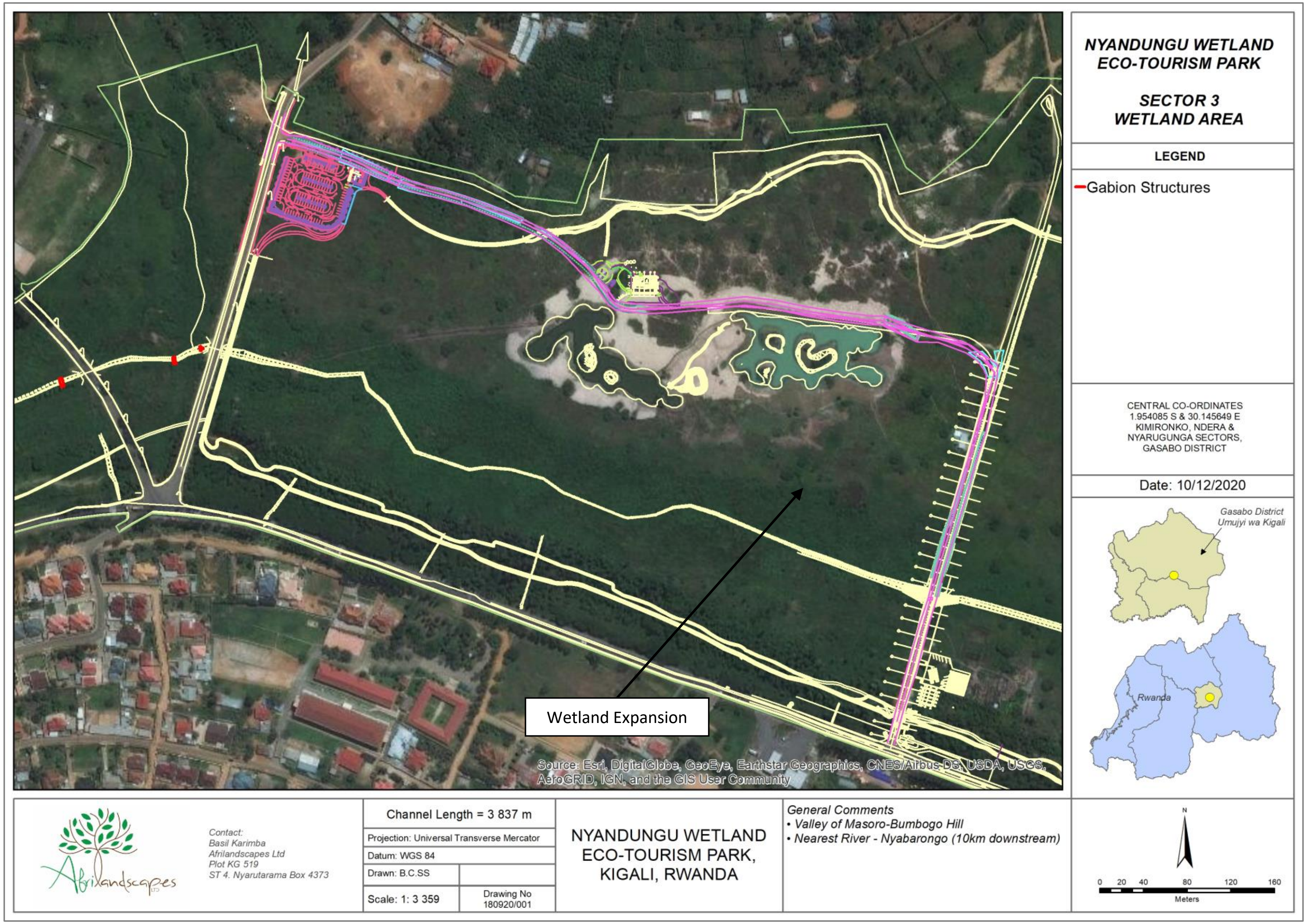

Figure 12 **Wetland area of Sector 3 of Nyandungu wetland eco-tourism park**

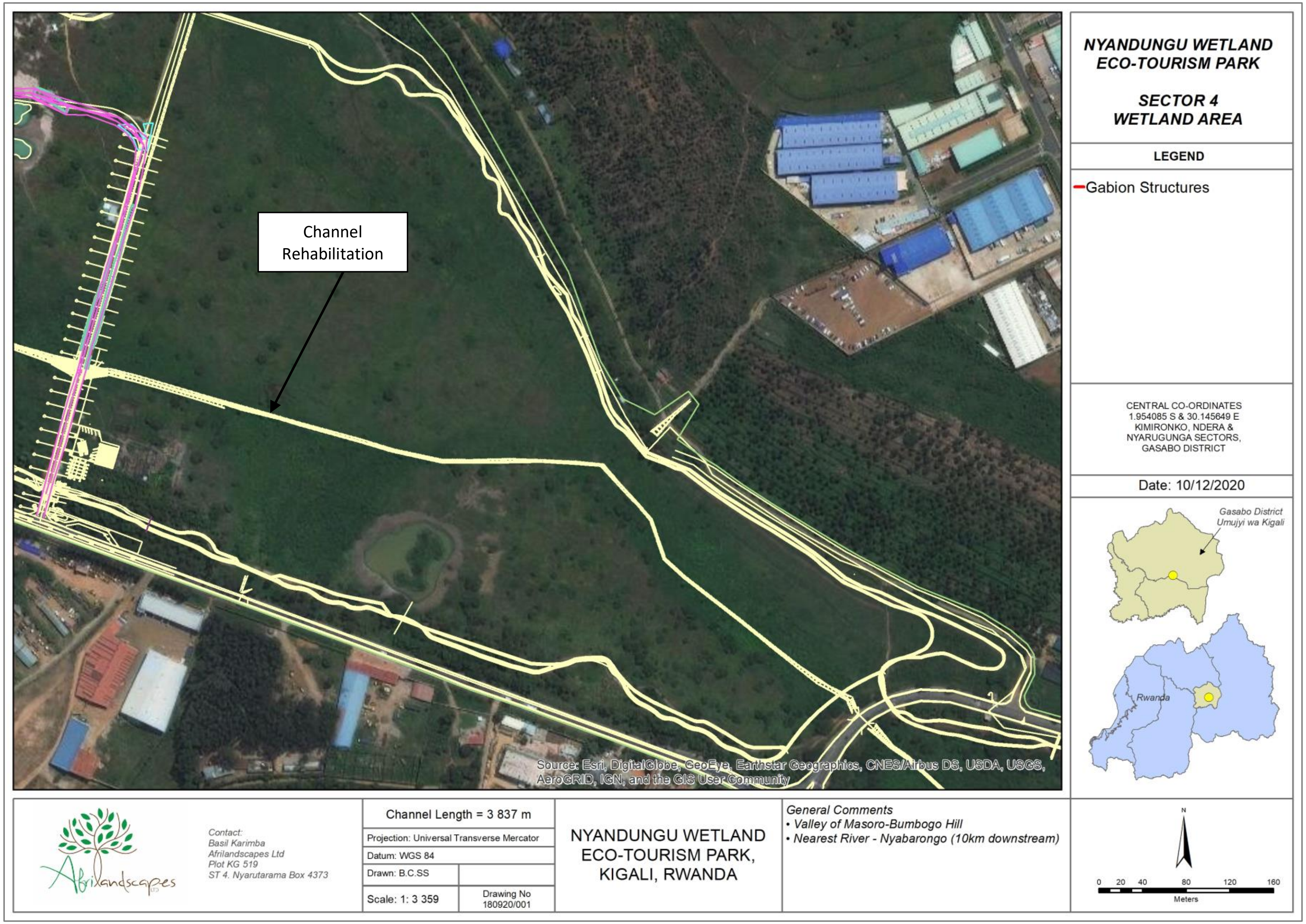

Figure 13 **Wetland area of Sector 4 of Nyandungu wetland eco-tourism park**

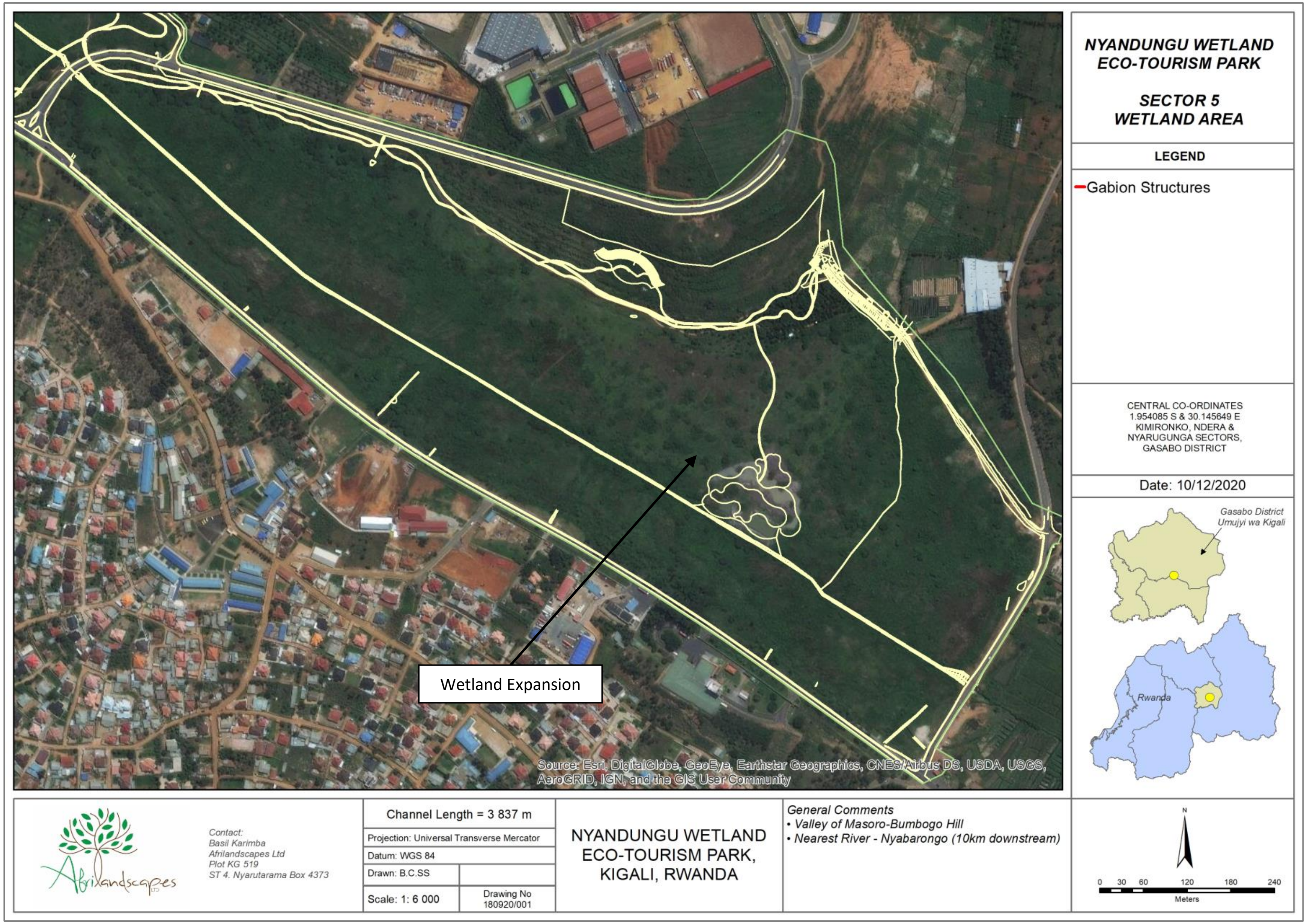

Figure 14 **Wetland area of Sector 5 of Nyandungu wetland eco-tourism park**

5.3 Present Ecological State (PES)

The wetlands were all assessed based on a pre-rehabilitation and post- rehabilitation state. It must be noted that rehabilitation is not complete and the findings in this report are based on post- rehabilitation at the time of this report.

5.3.1 Index of Habitat Integrity for riparian areas

The Index of Habitat Integrity tool (Kleynhans, 1996) was used to determine the integrity of the riparian zone associated with the channeled valley bottom wetland (HGM 1). The pre-rehabilitation and post- rehabilitation assessments and have been provided in Tables 9 and 10. The pre-rehabilitation state was determined by assessments of the immediate surrounding areas. The results for the **pre-rehabilitation state show a PES category of E** (39, Table 9): ""The loss of natural habitat, biota and basic ecosystem functions are extensive." The riparian areas are heavily disturbed, invaded by alien plant species and the water quality has been compromised. The surrounding areas are largely transformed with numerous road crossings, industrial sites, dump sites and residential/informal encroachment.

The results for the **post-rehabilitation state show a PES category of D** (59.6, Table 10): "Largely modified. A large loss of natural habitat, biota and basic ecosystem functions have occurred." Although the score is still low, there has been a significant improvement with channel/flow modification, indigenous vegetation and water quality. Given that the catchment is so transformed, the riparian area is still heavily impacted upon.

Table 9 Pre-rehabilitation PES score using the Index of Habitat Integrity tool (Kleynhans, 1999) for HGM Unit 1

Riparian Zone					
Criterion	**Score**	**Weighting**	**Actual**	**Potential**	
Indigenous vegetation removal	22	13	286	325	
Exotic vegetation encroachment	23	12	276	300	
Bank Erosion	8	14	112	350	
Channel modification	14	12	168	300	
Water abstraction	10	13	130	325	
Inundation	8	11	88	275	
Flow modification	14	12	168	300	
Water quality	22	13	286	325	
Totals			**1514**	**2500**	**60.56**
Category					**39.44**

Table 10 Post- rehabilitation PES score using the Index of Habitat Integrity tool (Kleynhans, 1999) for HGM Unit 1

Riparian Zone					
Criterion	**Score**	**Weighting**	**Actual**	**Potential**	
Indigenous vegetation removal	12	13	156	325	
Exotic vegetation encroachment	10	12	120	300	
Bank Erosion	6	14	84	350	
Channel modification	9	12	108	300	
Water abstraction	6	13	78	325	
Inundation	10	11	110	275	
Flow modification	10	12	120	300	
Water quality	18	13	234	325	
Totals			**1010**	**2500**	**40.4**
Category					**59.6**

5.4 Index of Habitat Integrity for riparian areas

The pre-rehabilitation state of the primary wetland system was very poor. Some of the key services that this type of wetland provides did not score highly (Figure 15). This is mostly due to historical transformation. The rehabilitated wetland state provides increased services in flood attenuation, stream-flow regulation, toxicant removal, tourism and education (Figure 16). Furthermore, roads and paths intersect these wetlands resulting in concentrated flow paths and a slight increase in compaction.

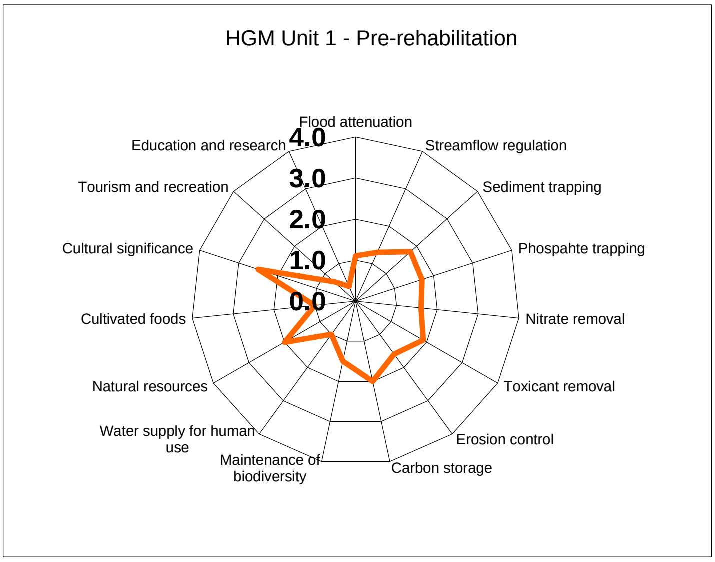

Figure 15 **Pre-rehabilitation WET-EcoServices of HGM Unit 1 of Sector 5 of Nyandungu wetland eco-tourism park**

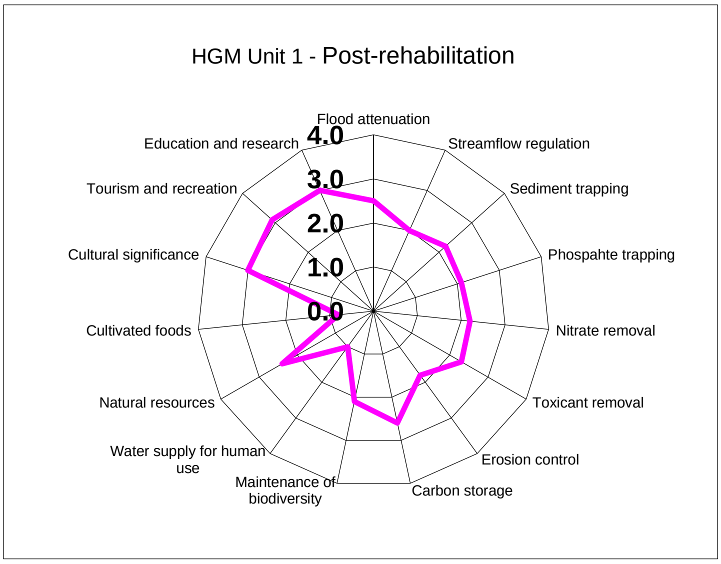

Figure 16 **Post-rehabilitation WET-EcoServices of HGM Unit 1 of Sector 5 of Nyandungu wetland eco-tourism park**

5.4.1 WET-Health (Macfarlane et al., 2008) of wetlands

A WET-Health assessment was undertaken for the Nyandungu wetland. Wetlands that are part of the same system but have split due to roads/dams were grouped together in the health assessments. The pre-rehabilitation state was based on historical data, soil evidence and surrounding systems.

- *Hydrology*

The valley bottom wetlands on-site were partially encroached by urban/commercial/ industrial areas, agriculture and modified historically. The present hydrological state of wetlands prior to rehabilitation was given a score of D, meaning that t**he impact of the modifications is clearly detrimental to the hydrological integrity. Approximately 50% of the hydrological integrity has been lost.** After the current rehabilitation, the wetland state was given a score of C, meaning that **the impact of the modifications on the hydrological integrity is clearly identifiable, but limited.** The MAP: PET ratio indicates that the wetlands are not dependant on direct precipitation falling onto the wetland, depending on flow from upstream or lateral seepage to a greater extent, making these wetlands more vulnerable to reduced flows.

The key factors influencing hydrological impacts on the wetland is the change in flow regime. There was some post- rehabilitation change in the overall health.

Table 11 The hydrology module associated with valley bottom wetland

Hydrology module	Valley Bottom (Pre-rehabilitation)	Valley Bottom (Post-rehabilitation)
Extent of the wetland (ha)	65	73
MAP:PET	0.4 – 0.49	0.4 – 0.49
Vulnerability factor	0,9	0,9
Combined score for increased and decreased flows	7.3	-2.0
Intensity of impact of factors potentially altering flow patterns	4 – Large	3 – Moderate
Magnitude of impact of canalisation and stream modification	0.4	0.1
Magnitude of impact of impeding features	2	2
Magnitude of impact of altered surface roughness	0,1	0
Impact of direct water losses	1,60	2.5
Magnitude of impact of recent deposition, infilling or excavation	1	0.44
Combined magnitude of impact of on-site activities	5.8 – Large	2.8
Combined magnitude score as a result of impacts on hydrological functioning	7	5
Overall hydrological health	The impact of the modifications is clearly detrimental to the hydrological integrity. Approximately 50% of the hydrological integrity has been lost.	The impact of the modifications on the hydrological integrity is clearly identifiable, but limited.
Present hydrological state of the HGM unit	**D**	**C**
Trajectory of change of wetland hydrology	(→)	(→)

- *Vegetation*

The present state of wetland vegetation of the valley bottom wetlands has been given a class of B as the vegetation composition has been significantly improved. Many areas surrounding the original wetland area had been historically disturbed and encroached by agriculture. However, as a buffer of indigenous vegetation including the wetland vegetation, a drastic improvement has occurred.

Table 12 Vegetation module associated with valley bottom wetland

Vegetation module	Valley Bottom (Pre-rehabilitation)	Valley Bottom (Post-rehabilitation)
Extent of the HGM unit (ha)	65	73
Identify and estimate the extent of each disturbance class	Large	Small
Magnitude of impact score	6.3	2.0
Present vegetation state	D	B
Trajectory of change to wetland vegetation	(→)	(→)
Overall vegetation health	Vegetation composition has been substantially altered but some	A very minor change to vegetation composition is evident at the site.

	characteristic species remain, although the vegetation consists numerous introduced, alien and/or ruderal species.	
Alien vegetation present (%)	35	15

- *Geomorphology*

The overall geomorphological health of the valley bottom wetlands was classified as D before and after rehabilitation occurred, which is largely modified. This was due to existing deposition and road intersections. The trajectory of change if the impacts do not continue is likely to remain stable (→).

Table 13 Geomorphology module for wetlands associated with the Channeled VB and Un-channeled VB wetlands

Geomorphology module	Valley Bottom (Pre-rehabilitation)	Valley Bottom (Post-rehabilitation)
Extent of the HGM unit (ha)	65	73
Impacts of channel straightening	1.5	1.5
Extent of impact of infilling	30	30
Impacts of changes in runoff characteristics	1.1	1.1
Impacts of erosion	0.2	0.2
Impacts of deposition	1.5	1.5
Present geomorphic state	**D**	**D**
Trajectory of change of geomorphic state	(→)	(→)
Overall geomorphological health	**Largely modified. A large change in geomorphic processes has occurred and the system is appreciably altered.**	**Largely modified. A large change in geomorphic processes has occurred and the system is appreciably altered.**

- *Overall Health*

The overall health based on the combined impact score for the pre-rehabilitation state is D (Largely modified). However, due to the current interventions, the overall health is now a C (moderately modified) - moderate change in ecosystem processes and loss of natural habitats has taken place but the natural habitat remains predominantly intact. The primary impact upon this system is the impact of surrounding developments on the water quality and flow regimes of these systems.

As a result of the interventions, there will likely be a change in the flow regime (attenuation of floods, improvement in water quality etc.).

6. POTENTIAL IMPACT PREDICTIONS AND DESCRIPTIONS

The site is in a visibly modified condition. The primary surrounding impacts are commercial farming which has led to reductions in seasonal hydrological functioning. The geomorphology is in a modified state due to some historical trenching. The actual site was historically transformed for many years. However, the surrounding natural vegetation is intact around the watercourse areas.

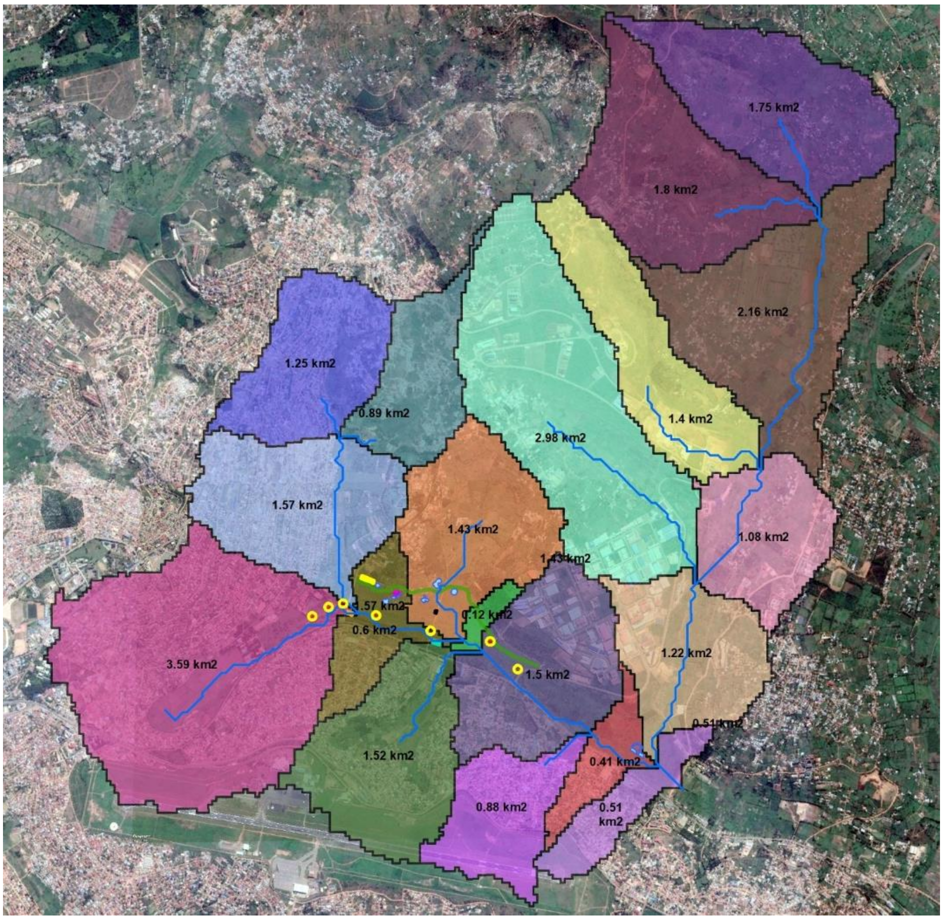

Figure 17 Catchment of Nyandungu wetland eco-tourism park

6.1 Existing Impacts

The major drivers of the identified impacts are the change in surface cover (impervious surfaces). These impacts are described further in the points below.

1. **Impact on local hydrology and riparian areas:** by disturbance of movement of water/flow towards the wetland and its soils, increased runoff due to hardened surfaces and reduction of water quality through:

    - A change from pervious areas to impervious areas (plots, buildings and roads) resulting in increased runoff and the likely transport of sediment into the river;
    - Altered water flows into the wetland linked to catchment changes;
    - An increase in the turbidity of the stream will occur as a result of the sedimentation. Contaminants from spoil sites may infiltrate the stream leading to a reduction in water quality. The placement of the spoil site is likely to be invaded by alien vegetation which are, by nature, excellent pioneer species.

2. **Change in the land cover and roughness characteristics:** by the removal of vegetation and topsoil, addition of spoil sites leading to wash, and compaction by heavy machinery resulting in increased runoff. This impact is linked to subsequent changes in the hydrological partitions and modifications to the slope and soil characteristics (changes to vegetation cover, root content and infiltration rates). Each of these are further described as:

- The potential increase in slope (berm banks) with the addition of spoil sites and bank construction will enhance existing erosion potential (greater energy for sediment wash);
- The reduction in vegetation cover will open bare soil therefore reducing the surface roughness and increasing the erosive potential to the elements (wind and rain). Sheet wash, rill and gully erosion will be likely and may lead to the collapse or slumping of bank areas that would lead to burying marginal and edge instream habitat. A reduction in active root mass, which bind soil particles will also be a direct impact of this.
- An increase in compaction of the soils along the edge of the site where heavy machinery traverse (access routes and temporary roads) will lead to an increase in the runoff.

3. **Loss of natural/indigenous vegetation and alien invasion:** clearing of riparian vegetation in and around the wetland area and change in the composition and structure of the riparian vegetation. This is caused by:

   - Direct removal of indigenous species (limited as site has transformed and is dense in alien vegetation) for construction clearing.
   - The removal of indigenous species and further ruderal and alien plant invasion will occur. This may also further increase the rate of erosion as discussed previously.
   - The likely introduction of alien seeds in spoil obtained from nearby areas.
   - The removal of top soil will further encourage alien pioneers which can grow under such conditions.
   - Certain alien plant species are known to alter the pH of the soil and groundwater in the vicinity of these plants. Furthermore a loss in biodiversity due to the often poisonous or undesirable taste of these plants is likely.

4. **Pollution (water, air and noise):** This will be as a result of:

   - An increase in pollution due to air emissions from heavy machinery, potential leaks of fuel, grease and oil from the heavy machinery. Wash related to the above-mentioned changes during rainfall events will lead to the movement of these waste substances into the soil and the streams.
   - The storage of hazardous chemicals such as fuel, ablution facilities that may leak, and any spills, such as concrete, during construction. If not properly managed, this may lead to impacts long after the construction is completed.
   - Aquatic species may be poisoned thus resulting in death (in the localised area) or movement away from the site. This may influence the ability of the aquatic species from breeding successfully as the lack of breeding species may disrupt upstream and downstream populations. Changes to Dissolved Oxygen (DO) and Total Dissolved Solids (TDS) as a result of a reduction in aquatic plants may detrimentally impact on invertebrate and vertebrate species. However, these impacts are very unlikely.
   - Noise pollution by machinery and labourers may affect nearby residents and wildlife. Fauna that would otherwise cross through this system may avoid this area resulting in no dispersal of seeds and a lack in local biodiversity.
   - Air pollution from the use of machinery (heavy vehicles/machines and generators) may result is a poor quality of life during this period for nearby households. Furthermore, bird populations may be negatively impacted upon.

7. IMPACT SIGNIFICANCE SCORING

The score of significance of pre-rehabilitation negative impacts before interventions is **8**. The full results can be seen in Table 14. This depicts an impact "*material / important to investigate - the impact is of importance and is therefore considered to have a substantial impact. Mitigation is required to reduce the negative impacts and such impacts need to be evaluated carefully*".

Through the current site rehabilitation measures, a reduction in the in the significance of the impacts has occurred. A score of **7.88** (Table 15) indicates that post-rehabilitation, the impacts are Marginal / slight / minor - The impact is of little importance, but may require limited mitigation; or it may be rendered acceptable in light of proposed mitigation.

Table 14 Significance of impacts for the pre-rehabilitation state

Site		Nature of potential impacts	Spatial extent	Severity / intensity / magnitude	Duration	Resource loss	Reversibility	Probability	SIGNIFICANCE SCORE
Total coverage (hardened surfaces)	1	Potential for soil erosion from increased stormwater runoff	3	2	3	4	3	0.2	3
	2	Potential for pollutants from increased stormwater runoff	3	2	3	4	3	0.2	3
	3	Potential for increased flooding	3	1	3	3	3	0.2	2.6
Internal roads	4	Potential for soil erosion from increased stormwater runoff	3	3	3	4	3	0.2	3.2
	5	Potential for pollutants from increased stormwater runoff	3	3	3	4	3	0.3	4.8
	6	Traffic, security, noise, dust, activity, air quality	1	3	2	4	3	0.3	3.9
	7	Activity on site	2	3	3	4	7	0.9	17.1
Activities Sand mining, dumping, farming, urban encroachment	8	Increased traffic	3	3	3	4	7	0.6	12
	9	Security risks	2	3	7	4	7	0.3	6.9
	10	Increased noise	3	3	3	4	7	0.6	12
	11	Increased dust	3	3	3	4	7	0.6	12
	12	Decreased air quality	3	3	3	4	7	0.2	4
	13	Disturbance of biological corridor	3	3	7	5	7	0.7	17.5
	14	Change in flow regimes due to impervious areas and changes to slope	4	3	3	5	7	0.7	15.4
	15	Change of views – rural / peri-urban to urban	2	3	6	3	7	0.6	12.6
	16	Impact on aesthetics / sense of place	2	3	5	3	7	0.6	12
							Subtotal Negative		**142**
OVERALL NEGATIVE IMPACT OLD LAYOUT									**8.9**

Table 15 Significance of impacts for the post-rehabilitation state

Site		Nature of potential impacts	Spatial extent	Severity / intensity / magnitude	Duration	Resource loss	Reversibility	Probability	SIGNIFICANCE SCORE
Total coverage (hardened surfaces)	1	Potential for soil erosion from increased stormwater runoff	2	2	2	4	3	0.2	**2.6**
	2	Potential for pollutants from increased stormwater runoff	2	2	2	3	3	0.2	**2.4**
	3	Potential for increased flooding	3	1	3	3	3	0.2	**2.6**
Internal roads	4	Potential for soil erosion from increased stormwater runoff	2	3	2	3	3	0.2	**2.6**
	5	Potential for pollutants from increased stormwater runoff	2	2	3	3	3	0.3	**3.9**
	6	Traffic, security, noise, dust, activity, air quality	1	2	2	3	3	0.3	**3.3**
	7	Activity on site	2	3	3	4	7	0.9	**17.1**
Proposed activities Eco-tourism park (restaurant, cycle routes etc.) Increased wetland area Ponds/lakes	8	Increased traffic	3	3	3	4	7	0.6	**12**
	9	Security risks	2	3	4	4	7	0.3	**6**
	10	Increased noise	3	3	3	3	7	0.6	**11.4**
	11	Increased dust	2	2	2	4	7	0.6	**10.2**
	12	Decreased air quality	2	2	3	4	7	0.2	**3.6**
	13	Disturbance of biological corridor	3	2	3	3	7	0.7	**12.6**
	14	Change in flow regimes due to impervious areas and changes to slope	2	3	2	5	4	0.7	**11.2**
	15	Change of views – rural / peri-urban to urban	2	3	6	3	7	0.6	**12.6**
	16	Impact on aesthetics / sense of place	2	3	5	3	7	0.6	**12**
							Subtotal Negative		**126.1**
OVERALL NEGATIVE IMPACT OLD LAYOUT									**7.88**

8. SURFACE WATER MONITORING PROGRAMME

General

The mitigation measures and associated recommendations are:

- There is some potential for erosion. Measures should be taken to ensure that this is minimized where possible. Particularly around the earthen wall and upstream areas that may be subjected to head cutting.
- Floods may have an impact on the structures. Debris and silt should be cleared from flow infrastructure and the dams.
- Stormwater runoff should be managed around the site and structures (e.g. restaurant).
- The water stored in the dams should be used in a sustainable manner to reduce the impacts of waterlogging and desiccation when flows are low, sustainable use will ensure a stable ecosystem is progressively established and dam impacts reduced.
- If the alien species have become established during the construction periods then these must be removed and indigenous species planted.

Water Quality

In the instance of a spill on site the following procedure must be followed:

1. Minimization of contaminated areas and reuse of dirty water;
2. In the event of a severe storm, operation activities should be temporarily stopped to avoid spills and potential contamination;
3. Locate the source of the spill or overflow;
4. Stop the spill and prevent further spreading;
5. Spills from chemical containers must be contained within the storage shed;
6. Spills from machinery or pipes (including in the event of a peak storm event) should be checked regularly to avoid contamination and reduce the loss of clean water;
7. Spills from vehicles must be contained within the concreted site area and prevented from spreading;
8. Spilled petrochemicals can then be cleaned up and removed using the appropriate oil sponge, absorbent or spill kit (e.g. DriZit);
9. The spill must be reported to the site manager / supervisor and ECO; and
10. Depending on the significance of the spill, the incident may also need to be reported to the DEDTEA and DWS.

It is imperative that the conditions of the WMP are incorporated into work plans to ensure that all development sectors are aware of the requirements. Follow-up assessments are key to ensuring the success of the intervention measures and the tourism associated structures.

9. CONCLUSION

The work undertaken for this report provides information on the general hydrological characteristics, the differences between the current state and the historical state as well as general impacts including those on downstream users.

The results show that due to the on-going interventions, there has been a significant improvement in the overall state of the wetland park. Given the historical past of this wetland (small scale agriculture and canalisation), which resulted in less sustainable yield and greater flood risk, the rehabilitation of the wetland and inclusion of the dams has improved the sustainable yield (greater assurance of flow) and reduced the flood risk by attenuating peak events.

10. REFERENCES

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ANNEXURE A **Classification structure for inland systems up to Level 4**

<table>
<tr><th colspan="3">WETLAND / AQUATIC ECOSYSTEM CONTEXT</th></tr>
<tr><th>LEVEL 1:
SYSTEM</th><th>LEVEL 2:
REGIONAL SETTING</th><th>LEVEL 3:
LANDSCAPE UNIT</th></tr>
<tr><td rowspan="4">Inland Systems</td><td rowspan="4">DWA Level 1 Ecoregions
OR
NFEPA WetVeg Groups
OR
Other special framework</td><td>Valley Floor</td></tr>
<tr><td>Slope</td></tr>
<tr><td>Plain</td></tr>
<tr><td>Bench
(Hilltop / Saddle / Shelf)</td></tr>
</table>

<table>
<tr><th colspan="3">FUNCTIONAL UNIT</th></tr>
<tr><th colspan="3">LEVEL 4:
HYDROGEOMORPHIC (HGM) UNIT</th></tr>
<tr><td>HGM type</td><td>Longitudinal zonation/ Landform / Outflow drainage</td><td>Landform / Inflow drainage</td></tr>
<tr><td>A</td><td>B</td><td>C</td></tr>
<tr><td rowspan="18">River (Channel)</td><td rowspan="2">Mountain headwater stream</td><td>Active channel</td></tr>
<tr><td>Riparian zone</td></tr>
<tr><td rowspan="2">Mountain stream</td><td>Active channel</td></tr>
<tr><td>Riparian zone</td></tr>
<tr><td rowspan="2">Transitional stream</td><td>Active channel</td></tr>
<tr><td>Riparian zone</td></tr>
<tr><td rowspan="2">Upper foothill rivers</td><td>Active channel</td></tr>
<tr><td>Riparian zone</td></tr>
<tr><td rowspan="2">Lower foothill rivers</td><td>Active channel</td></tr>
<tr><td>Riparian zone</td></tr>
<tr><td rowspan="2">Lowland river</td><td>Active channel</td></tr>
<tr><td>Riparian zone</td></tr>
<tr><td rowspan="2">Rejuvenated bedrock fall</td><td>Active channel</td></tr>
<tr><td>Riparian zone</td></tr>
<tr><td rowspan="2">Rejuvenated foothill rivers</td><td>Active channel</td></tr>
<tr><td>Riparian zone</td></tr>
<tr><td rowspan="2">Upland floodplain rivers</td><td>Active channel</td></tr>
<tr><td>Riparian zone</td></tr>
<tr><td>Channelled valley-bottom wetland</td><td>(not applicable)</td><td>(not applicable)</td></tr>
<tr><td>Unchannelled valley-bottom wetland</td><td>(not applicable)</td><td>(not applicable)</td></tr>
<tr><td rowspan="2">Floodplain wetland</td><td>Floodplain depression</td><td>(not applicable)</td></tr>
<tr><td>Floodplain flat</td><td>(not applicable)</td></tr>
<tr><td rowspan="6">Depression</td><td rowspan="2">Exorheic</td><td>With channelled inflow</td></tr>
<tr><td>Without channelled inflow</td></tr>
<tr><td rowspan="2">Endorheic</td><td>With channelled inflow</td></tr>
<tr><td>Without channelled inflow</td></tr>
<tr><td rowspan="2">Dammed</td><td>With channelled inflow</td></tr>
<tr><td>Without channelled inflow</td></tr>
<tr><td rowspan="2">Seep</td><td>With channelled outflow</td><td>(not applicable)</td></tr>
<tr><td>Without channelled outflow</td><td>(not applicable)</td></tr>
<tr><td>Wetland flat</td><td>(not applicable)</td><td>(not applicable)</td></tr>
</table>

Note: 2nd row of Table provides the criterion for distinguishing between wetland units in each column

ANNEXURE B Wetland and soil classification field datasheet example

Sampling Sheet Summary	
Wetland	kigali
Area (ha)	72
Indicator	Soil and vegetation
Connectivity (level 1)	Inland
Eco region (level 2)	South Eastern Uplands
Landscape setting (level 3)	Riparian system
HGM Type (level 4A)	Endhoreic
Longitudinal zonation (level 4B)	With channel
Hydrological regime	Frequent Inundation
Soil characteristics	Hue – Gley 2 to 5YR Value – 4 Chroma – 2 (Dark Reddish Gray) Depth sampled: 0-0.5m
Comment	No change in soil characteristics

ANNEXURE C Steps for Riparian Delineation

Steps for Riparian Delineation in the field

To delineate riparian areas, use the terrain unit indicator, vegetation indicator species, soil wetness indicator, combined with

- Geomorphology of the banks; and
- Extent of riparian vegetation.

Evidence of alluvial deposits can also be used.

STEPS to delineating the riparian zone:

I. Is the site relatively undisturbed (banks have not been extensively engineered, and the site is predominantly indigenous, naturally occurring vegetation)? If yes, proceed to step II. If no, proceed to step V.

II. Starting at the edge of the channel, use the regional riparian vegetation indicator list, identify the edge of the zone of (obligate) riparian plants.

III. At this point, check:

a. If there are any hydric indicators in the soil (refer to Wetland Delineation component).
b. If you are still in a zone of unconsolidated recent alluvial sediment.

If yes for either **a** or **b**, proceed outwards from the channel to identify the edge of these zones.

Once the answer to **a** and **b** are no, follow the same steps (II and III) using preferential and/or facultative riparian plant species *(Refer to the steps 1 to 12 from the vegetation assessment section below for further detail).*

Following completion of the above, proceed to step IV.

IV. Examine the geomorphology (shape) of the channel and banks. After moving away from the channel during steps II and III, you should be at or close to the edge of the top of the "macro-channel" bank (in the case of erosive rivers) or the edge of the active floodplain or flood zone (in the case of alluvial depositional rivers). At, or close to, this point you should see an inflection point (change in slope) between the riparian area and the upland (terrestrial) slopes. This can be taken as the edge of the riparian zone.

Using Reference Sites:

V. For sites which have been heavily disturbed (i.e. where there is almost no indigenous vegetation remaining, and/or where the banks have been heavily engineered such that it is no longer possible to identify the original morphology of the banks), then a REFERENCE site will need to be located. The Reference site will need to be close by on the same or a similar sized river system, in an area of similar topography. The Reference Site can be used to provide an indication of the likely riparian extent prior to disturbance. Once the reference site is located, proceed with step II.

Where problems may be encountered:

On floodplains, it is important to check whether the floodplain is active (i.e. regularly flooded under the current climatic regime) or a relict floodplain (meaning that the floodplain depositional area formed due to a wetter historical climate and now is no longer regularly flooded). The type of vegetation on the floodplain surface, presence of soil wetness indicators and the presence of ox-bows and other riparian and wetland features would provide the indications of the current levels of flooding/inundation/saturation.

ANNEXURE D Significance Scoring Method[1]

1. Nature of impact

The environmental impacts of a project are those resultant changes in environmental parameters, in space and time, compared with what would have happened had the project not been undertaken. It is an appraisal of the type of effect the proposed activity would have on the affected environmental parameter. Its description should include what is being affected, and how.

2. Spatial extent

This addresses the physical and spatial scale of the impact. A series of standard terms relating to the spatial extent of an impact / effect are outlined in Table 1.

Table 1 Rating scale for the assessment of the spatial extent of predicted effect / impact.

Rating	Spatial descriptor
7	International - The impacted area extends beyond national boundaries
6	National - The impacted area extends beyond provincial boundaries
5	Ecosystem - The impact could affect areas essentially linked to the property in terms of significantly impacting ecosystem functioning
4	Regional - The impact could affect a subset of the ecosystem, extending over more than one landscape
3	Landscape - The impact could affect all areas generally visible to the naked eye, as well as those areas essentially linked to the property in terms of ecosystem functioning
2	Site related - The impacted area extends further than the actual physical disturbance footprint; the impact could affect the whole, or a measurable portion of a number of properties, forming part of the development area or within close proximity of the development property
1	Local - The impacted area extends only as far as the activity e.g. a footprint; the loss is considered inconsequential in terms of the spatial context of the relevant environmental aspect

3. Severity / Intensity / Magnitude

A qualitative assessment of the severity of a predicted impact / effect is undertaken. Quantitative measures are undertaken wherever possible. A series of standard terms relating to the magnitude of an impact / effect are outlined in Table 2.

Table 2 Rating scale for the assessment of the severity of a predicted effect / impact.

Rating	Magnitude descriptor
7	Total / consuming / eliminating - Function or process of the affected environment is altered to the extent that it is permanently changed in character / intrinsic value. Certainty of presence of conservation-important flora.
6	Profound / considerable / substantial - Function or process of the affected environment is altered to the extent where it is permanently modified to a sub-optimal state.

[1] adapted from Glasson J, Therivel R & Chadwick A. Introduction to Environmental Impact Assessment, 2nd Edition. 1999. pp 258. Spon Press, United Kingdom.

5	Material / important - Function or process of the affected environment is altered to the extent where some components of a natural system are permanently modified to a sub-optimal state.
4	Discernible / noticeable - The affected environment is altered, but overall function and process continue, albeit in a modified way. Uncertainty of presence of conservation-important flora.
3	Marginal / slight / minor - The affected environment is altered, but natural function and process continue.
2	Unimportant / inconsequential / indiscernible - The impact alters the affected environment in such a way that the natural processes or functions are negligibly affected.
1	No effect. Certainty that conservation-important flora are not present.

4. Duration

This describes the predicted lifetime of the impact.

Table 3 Rating scale for the assessment of the temporal scale of a predicted effect / impact.

Rating	Temporal descriptor
7	Long-term – Permanent. Beyond decommissioning and cannot be negated on decommissioning. More than 15 years.
3	Medium term – Lifespan of the project. Reversible over time. 5 to 15 years.
1	Short-term – Quickly reversible. Less than the project lifespan. The impact will either disappear with mitigation or will be mitigated through natural process in a span shorter than any of the phases. 0 to 5 years.

5. Irreplaceable loss of resources

Environmental resources cannot always be replaced; once destroyed, some may be lost forever. It may be possible to replace, compensate for or reconstruct a lost resource in some cases, but substitutions are rarely ideal. The loss of a resource may become more serious later, and assessment must take this into account.

Table 4 Rating scale for the assessment of the loss of resources due to a predicted effect / impact.

Rating	Resource loss descriptor
7	Long-term – The loss of a non-renewable / threatened resource which cannot be renewed / recovered with or through natural process, in a time span of over 15 years, or by artificial means.
5	Long-term – The loss of a non-renewable / threatened resource which cannot be renewed / recovered with or through natural process, in a time span of over 15 years, but can be mitigated by other means.
4	Loss of an 'at risk' resource - one that is not deemed critical for biodiversity targets, planning goals, community welfare, agricultural production, or other criteria, but cumulative effects may render such loss as significant.
3	Medium term – The resource can be recovered within the lifespan of the project. The resource can be renewed / recovered with mitigation or will be mitigated through natural process in a span between 5 and 15 years.
2	Loss of an 'expendable' resource - one that is not deemed critical for biodiversity targets, planning goals, community welfare, agricultural production, or other criteria.
1	Short-term – Quickly recoverable. Less than the project lifespan. The resource can be renewed / recovered with mitigation or will be mitigated through natural process in a span shorter than any of the phases, or in a time span of 0 to 5 years.

6. Reversibility / potential for rehabilitation

The distinction between reversible and irreversible impacts is a very important one, and the irreversible impacts, not susceptible to mitigation, can constitute significant impacts in an EIA (Glasson *et al*, 1999). The potential for rehabilitation is the major determinant factor when considering the temporal scale of most predicted impacts.

Table 5 Rating scale for the assessment of reversibility of a predicted effect / impact.

Rating	Reversibility descriptor
7	Long-term – The impact / effect will never be returned to its benchmark state.
3	Medium term – The impact / effect will be returned to its benchmark state through mitigation or natural processes in a span shorter than the lifetime of the project, or in a time span between 5 and 15 years.
1	Short-term – The impact / effect will be returned to its benchmark state through mitigation or natural processes in a span shorter than any of the phases of the project, or in a time span of 0 to 5 years.

7. Probability

An assessment of the probability of an impact / effect was undertaken in accordance with Table 6.

Table 6 Rating scale for the assessment of the probability of a predicted effect / impact.

Rating	Probability descriptor
1.0	Absolute certainty
0.9	Near certainty / very high probability
0.7 – 0.8	High probability – to be expected
0.4 - 0.6	Likelihood / normal anticipation – to be anticipated
0.3	Seriously anticipated possibility
0.2	Possibility
0.0 - 0.1	Remote possibility

8. Mitigation

The potential to mitigate the negative impacts and enhance the positive impacts should be determined for each identified impact, mitigation objectives that would result in a measurable reduction in impact should be provided. For each impact, practical mitigation measures that can affect the significance rating should be recommended. Management actions that could enhance the condition of the environment (i.e. potential positive impacts of the proposed project) should be identified. Where no mitigation is considered feasible, this must be stated and the reasons provided (DEAT, 2002).

The significance of environmental impacts will be assessed taking into account any proposed mitigations. The significance of the impact "without mitigation" is the prime determinant of the nature and degree of mitigation required.

Table 7 Significance scoring of negative impact / effect

Scoring value	Significance
35	Total / consuming / eliminating - In the case of adverse impacts, there is no possible mitigation that could offset the impact, or mitigation is difficult, expensive, time-consuming or some combination of these. Mitigation may not be possible / practical. Critically important natural asset lost.
26 - 34	Profound - In the case of adverse impacts, there are few opportunities for mitigation that could offset the impact, or mitigation has a limited effect on the impact. Social, cultural and economic activities of communities are disrupted to such an extent that their operation is severely impeded. Mitigation may not be possible / practical. Threatened natural asset lost..
21 – 25	Considerable / substantial - The impact is of great importance. Failure to mitigate with the objective of reducing the impact to acceptable levels could render the entire project option or entire project proposal unacceptable. Mitigation is therefore essential. At-risk natural asset lost.
8 – 20	Material / important to investigate - The impact is of importance and is therefore considered to have a substantial impact. Mitigation is required to reduce the negative impacts and such impacts need to be evaluated carefully.
5 – 7	Marginal / slight / minor - The impact is of little importance, but may require limited mitigation; or it may be rendered acceptable in light of proposed mitigation.
0 – 4	Unimportant / inconsequential / indiscernible; or it may be rendered acceptable in light of proposed mitigation.